

Notes: Gopinath et al. (QJE, 2017)

Capital Allocation and Productivity in South Europe

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1 Big picture

- **Question.** Why did southern European countries receive large capital inflows and yet experience weak productivity growth after the adoption of the euro? More specifically, can a decline in real interest rates *worsen* capital allocation and reduce aggregate productivity when firms face financial frictions?
- **Setting.** The paper combines new firm-level evidence for **Spanish manufacturing firms, 1999–2012** with a dynamic model of **heterogeneous firms, size-dependent borrowing constraints**, and **capital adjustment costs**. It then compares Spain with other euro-area countries.
- **Core challenge.** A decline in borrowing costs is normally thought to improve allocation by relaxing credit constraints. The paper argues that this intuition can fail in transition: lower real interest rates may channel capital toward firms with **high net worth** rather than firms with **high productivity**, raising dispersion in the return to capital and lowering aggregate efficiency.
- **Main contribution.** The paper does two things at once:
 1. it documents the **time-series evolution of misallocation** within a country using broad firm-level data, and
 2. it develops a tractable model that links those reallocations to **financial frictions and capital flows**.
- **Why it matters.** This paper turns the discussion of “southern European imbalances” into a micro-founded misallocation story. It implies that lower real interest rates can generate **capital inflows together with lower TFP**, not just more investment.
- **Main empirical question.** Between 1999 and 2012, did Spanish manufacturing experience a deterioration in the allocation of capital across firms, and can the decline in the real interest rate explain that pattern?
- **Main empirical facts.**
 - The dispersion of the log marginal revenue product of capital, log MRPK, rises strongly over time.
 - The dispersion of the log marginal revenue product of labor, log MRPL, is roughly flat.
 - TFP relative to its efficient level falls over time.
 - Capital becomes more dispersed across firms, and the correlation between firm productivity and firm capital declines.
- **Main model result.** A fall in the real interest rate can increase capital flows and aggregate capital accumulation while also **worsening the allocation of capital**. In the model, richer firms expand faster because they can finance investment earlier; poorer but productive firms lag behind because they must self-finance into the future.
- **Main quantitative result.** The baseline model with heterogeneous size-dependent financial frictions generates:
 - a decline in TFP relative to its efficient level of about **2.3 percentage points** between 1999 and 2007 versus **3.1 points** in the data,

- an increase in MRPK dispersion of about **0.074** versus **0.034** in the data,
- strong growth in capital and debt, with debt rising more than one-for-one with capital.
- **Cross-country takeaway.** Similar patterns of rising MRPK dispersion and falling allocative efficiency appear in **Italy** and **Portugal**, but not in **Germany**, **France**, or **Norway**. The authors interpret this as suggestive evidence that firms in the South operated in less well-developed financial markets.
- **Economic takeaway.** Lower real interest rates are not automatically productivity-enhancing. When financial frictions depend on firm balance sheets, they can fuel a **mismatch boom**: more capital enters the economy, but too much of it goes to the wrong firms.

2 Data and measurement (key objects)

2.1 ORBIS-AMADEUS sample and coverage

- **Main source.** The firm-level data come from **ORBIS-AMADEUS**, assembled from balance sheets, income statements, and profit-and-loss accounts.
- **Country and sector.** The main empirical analysis focuses on **Spanish manufacturing** firms from **1999 to 2012**.
- **Why this dataset matters.** About **99%** of firms in the data are **private**, so the sample is much broader than datasets built from listed firms alone.
- **Coverage.** Relative to Eurostat Structural Business Statistics, the sample covers roughly:
 - **65%–71%** of employment,
 - about **72%–75%** of the wage bill,
 - about **72%–79%** of gross output.
- **Cleaning strategy.** The authors merge multiple ORBIS and AMADEUS vintages to reduce missing-variable problems, reporting-lag problems, and the survival bias that arises when inactive reporters disappear from later vintages.

2.2 Representativeness by firm size

- **Size composition.** The sample is broadly representative of Spanish manufacturing by firm size.
- In 2006, firms with **1–19 employees** account for roughly **19%** of the wage bill in ORBIS-AMADEUS versus **20%** in Eurostat.
- Firms with **20–249 employees** account for roughly **47%** of the wage bill in ORBIS-AMADEUS versus **43%** in Eurostat.
- Firms with **250+ employees** account for roughly **34%** of the wage bill in ORBIS-AMADEUS versus **37%** in Eurostat.
- **Interpretation.** The sample is especially useful because it captures small and medium-sized firms, which are central in European manufacturing and in the paper’s mechanism.

2.3 Industry, firm, and sample definitions

- **Industry definition.** Industries are defined at the **four-digit NACE Rev. 2** level.
- **Main samples.**
 - The **full sample** includes firms that enter and exit the data over time.
 - The **permanent sample** keeps firms continuously present in the panel.
- **Why both samples are useful.** The permanent sample isolates within-firm dynamics and avoids composition effects from entry and exit; the full sample comes closer to the population-level Eurostat coverage.

2.4 Measured firm-level objects

- **Nominal value added.** Measured as operating revenue minus materials.
- **Real output.** Nominal value added divided by a **two-digit industry output price deflator**.
- **Labor input.** Measured by the **wage bill** deflated by the same output price deflator. Using the wage bill instead of employment partly controls for worker quality differences across firms.
- **Capital input.** Measured by the **book value of fixed assets**, deflated with the price of investment goods. The authors include both tangible and intangible fixed assets in their baseline capital measure.
- **Firm productivity.** The paper often works with a revenue-based productivity concept that combines physical productivity and demand. In the static measurement framework this is denoted $\tilde{Z}_{ist}$; in the dynamic model it is denoted Z_{it} .

2.5 Misallocation objects

- **MRPK.** The marginal revenue product of capital.
- **MRPL.** The marginal revenue product of labor.
- **TFPR.** Revenue-based total factor productivity, which captures wedges and markups.
- **TFP.** Industry output relative to an aggregate of industry capital and labor.
- **Efficient TFP.** The counterfactual TFP^e that would obtain if MRPK, MRPL, and TFPR were equalized across firms within an industry.

2.6 Key descriptive facts

- **Dispersion fact.** MRPK dispersion rises sharply from 1999 onward, with an acceleration after 2008.
- **Labor fact.** MRPL dispersion does **not** show a comparable upward trend.
- **Allocation fact.** TFP relative to its efficient level declines by roughly **3 percentage points** in the permanent sample and **7 percentage points** in the full sample between 1999 and 2007; by 2012 the cumulative losses are larger.

- **Capital allocation fact.** The standard deviation of log capital rises over time, while the cross-sectional correlation between firm productivity and firm capital falls.
- **Interpretation.** The economy is not simply seeing more heterogeneity in all factor returns. Rather, something specific is happening to **capital allocation**.

2.7 Unobservables and timing

- **Permanent productivity.** Firms differ in a permanent productivity component z_i^P .
- **Transitory productivity.** Firms face a persistent idiosyncratic shock z_{it}^T following an AR(1) process.
- **Aggregate driver.** The main transition experiment feeds into the model the observed decline in the **real interest rate** that began in the mid-1990s.
- **Key state variables.** Firm net worth, capital, permanent productivity, transitory productivity, and the real interest rate.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static Hsieh–Klenow measurement framework

Industry output is a CES aggregate of firm-level outputs:

$$Y_{st} = \left(\sum_{i=1}^{N_{st}} D_{ist} y_{ist}^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}. \quad (1)$$

Firm output is Cobb–Douglas:

$$y_{ist} = A_{ist} k_{ist}^\alpha \ell_{ist}^{1-\alpha}. \quad (2)$$

The firm chooses price, capital, and labor to maximize

$$\pi_{ist} = (1 - \tau_{ist}^y) p(y_{ist}) y_{ist} - (1 + \tau_{ist}^k) (r_t + \delta_{st}) k_{ist} - w_{st} \ell_{ist}. \quad (3)$$

The first-order conditions imply

$$\text{MRPL}_{ist} := \frac{1 - \alpha}{\mu} \frac{p_{ist} y_{ist}}{\ell_{ist}} = \frac{1}{1 - \tau_{ist}^y} w_{st}, \quad (4)$$

$$\text{MRPK}_{ist} := \frac{\alpha}{\mu} \frac{p_{ist} y_{ist}}{k_{ist}} = \frac{1 + \tau_{ist}^k}{1 - \tau_{ist}^y} (r_t + \delta_{st}), \quad (5)$$

where $\mu = \varepsilon/(\varepsilon - 1)$ is the constant markup.

Revenue productivity is

$$\text{TFPR}_{ist} := p_{ist} A_{ist} = \frac{p_{ist} y_{ist}}{k_{ist}^\alpha \ell_{ist}^{1-\alpha}} = \mu \left(\frac{\text{MRPK}_{ist}}{\alpha} \right)^\alpha \left(\frac{\text{MRPL}_{ist}}{1 - \alpha} \right)^{1-\alpha}. \quad (6)$$

Interpretation. In the Hsieh–Klenow benchmark, efficient allocation means equalizing factor returns across firms within an industry. Dispersion in MRPK, MRPL, and TFPR is therefore read as evidence of distortions or wedges in resource allocation.

3.2 Why rising MRPK dispersion with flat MRPL dispersion is special

The paper derives two accounting identities:

$$\text{Var}(mrpk) = \text{Var}(tfpr) + (1 - \alpha)^2 \text{Var}\left(\log \frac{k}{\ell}\right) - 2(1 - \alpha) \text{Cov}\left(tfpr, \log \frac{k}{\ell}\right), \quad (7)$$

$$\text{Var}(mrpl) = \text{Var}(tfpr) + \alpha^2 \text{Var}\left(\log \frac{k}{\ell}\right) + 2\alpha \text{Cov}\left(tfpr, \log \frac{k}{\ell}\right). \quad (8)$$

Interpretation. These identities show why the data are informative. If MRPK dispersion rises while MRPL dispersion stays flat, the covariance between TFPR and the capital-labor ratio must be falling. Intuitively, capital is becoming more weakly aligned with productivity.

3.3 Industry TFP and efficient TFP

Industry-level productivity is

$$\text{TFP}_{st} = \frac{Y_{st}}{K_{st}^\alpha L_{st}^{1-\alpha}} = \left[\sum_i \left(Z_{ist} \frac{\text{TFPR}_{st}}{\text{TFPR}_{ist}} \right)^{\varepsilon-1} \right]^{\frac{1}{\varepsilon-1}}, \quad (9)$$

with

$$\text{TFP}_{st}^e = \left(\sum_i Z_{ist}^{\varepsilon-1} \right)^{\frac{1}{\varepsilon-1}}.$$

The misallocation loss is

$$\Delta_{st} = \log(\text{TFP}_{st}) - \log(\text{TFP}_{st}^e). \quad (10)$$

The paper estimates a revenue-based productivity object

$$\tilde{Z}_{ist} = \frac{(p_{ist} y_{ist})^{\frac{\varepsilon}{\varepsilon-1}} (P_{st} Y_{st})^{-\frac{1}{\varepsilon-1}}}{P_{st} k_{ist}^\alpha \ell_{ist}^{1-\alpha}}. \quad (11)$$

Interpretation. TFP^e is the hypothetical productivity level under frictionless within-industry allocation. If the observed TFP drifts farther below TFP^e , misallocation is worsening even if measured output itself is still growing.

3.4 A reduced-form link from capital allocation to MRPK dispersion

The paper emphasizes that

$$\text{Var}_i(\log \text{MRPK}_{ist}) = \gamma_1 \text{Var}_i(\log Z_{ist}) + \gamma_2 \text{Var}_i(\log k_{ist}) - \gamma_3 \text{Cov}_i(\log Z_{ist}, \log k_{ist}), \quad (12)$$

for positive coefficients $\gamma_1, \gamma_2, \gamma_3$.

Interpretation. Even holding fixed the dispersion of productivity itself, MRPK dispersion rises if capital becomes more dispersed or if capital becomes less correlated with productivity. This is exactly what the Spanish data show.

3.5 Dynamic firm problem

In the structural model, firms solve

$$\max_{\{c_{it}, b_{it+1}, x_{it}, \ell_{it}, p_{it}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t U(c_{it}), \quad (13)$$

with CRRA utility

$$U(c_{it}) = \frac{c_{it}^{1-\gamma} - 1}{1-\gamma},$$

and budget constraint

$$c_{it} + x_{it} + (1 + r_t)b_{it} + \frac{\psi(k_{it+1} - k_{it})^2}{2k_{it}} = p_{it}y_{it} - w_t\ell_{it} + b_{it+1}, \quad (14)$$

capital accumulation

$$k_{it+1} = (1 - \delta)k_{it} + x_{it}, \quad (15)$$

and production

$$y_{it} = Z_{it}k_{it}^{\alpha}\ell_{it}^{1-\alpha}.$$

Interpretation. Unlike the static Hsieh–Klenow setup, capital is chosen dynamically under uncertainty and adjustment costs. This means some MRPK dispersion is endogenous even without financial distortions; the question is how much additional dispersion comes from borrowing frictions.

3.6 Size-dependent borrowing constraint

The key financial friction is

$$b_{it+1} \leq \theta_0 k_{it+1} + \theta_1 \phi(k_{it+1}) = \theta(k_{it+1})k_{it+1}, \quad (16)$$

which can be rewritten as

$$k' \leq \lambda_0 a' + \lambda_1 \frac{\phi(k')}{a'} a' = \lambda(k', a') a'. \quad (20)$$

Interpretation.

- If $\lambda_1 = 0$, borrowing capacity is proportional to net worth in the standard way.
- If $\lambda_1 > 0$, **larger firms can lever more**. This is the paper's central innovation.

Why is this so important? Because when the real interest rate falls, firms with high net worth can expand quickly, become effectively unconstrained, and drive down their own MRPK. Poorer firms cannot. Dispersion rises between them.

3.7 Productivity process

Firm productivity is decomposed as

$$Z_{it} = Z_t^A z_i^P \exp(z_{it}^T),$$

with transitory productivity following

$$z_{it}^T = -\frac{\sigma^2}{2(1+\rho)} + \rho z_{it-1}^T + \sigma u_{it}^z, \quad u_{it}^z \sim N(0, 1). \quad (17)$$

Interpretation. Firms differ in both permanent and transitory productivity. The model lets idiosyncratic risk itself generate some dispersion, but then asks whether the observed Spanish time series requires **financial frictions that depend on firm size**.

3.8 Labor demand and the source of MRPK dispersion

Labor demand is

$$\ell = Z^{\frac{\varepsilon-1}{1+\alpha(\varepsilon-1)}} \mu^{-\frac{\varepsilon}{1+\alpha(\varepsilon-1)}} \left(\frac{w}{1-\alpha} \right)^{-\frac{\varepsilon}{1+\alpha(\varepsilon-1)}} k^{\frac{\alpha(\varepsilon-1)}{1+\alpha(\varepsilon-1)}}. \quad (21)$$

In the baseline, labor markets are competitive and MRPL is equalized across firms. The first-order condition for capital can be summarized as

$$\text{MRPK} = (r + \delta) + \text{terms from borrowing constraints, adjustment costs, and risk.} \quad (22)$$

Interpretation. The model isolates three sources of MRPK dispersion:

- binding borrowing constraints,
- capital adjustment costs,
- ex post risk because capital is chosen before productivity is fully realized.

Among these, the paper's main emphasis is on the first source, especially when that constraint becomes weaker with firm size.

3.9 Real interest rate experiment and calibration

The real interest rate path is taken from Spanish corporate lending rates minus expected inflation. The model starts from a stochastic steady state with $r = 0.10$ before 1994 and is then fed the observed decline in the real rate.

Baseline parameter values are:

$$\begin{aligned} \beta &= 0.87, \quad \gamma = 2.00, \quad \delta = 0.06, \quad \varepsilon = 3.00, \quad \alpha = 0.35, \\ \rho &= 0.59, \quad \sigma = 0.13, \quad \pi = 0.80, \quad z_L = 0.785, \quad z_H = 1.86, \\ \psi &= 3.20, \quad \lambda_0 = 0.98, \quad \lambda_1 = 0.047. \end{aligned}$$

The last three parameters are calibrated to match three moments:

- the within-firm sensitivity of capital growth to productivity,
- the fraction of firms that borrow, about **0.90**,
- the positive cross-sectional relationship between leverage and firm size.

Interpretation. This is not a pure calibration-to-aggregates exercise. The model is disciplined by **micro moments** that directly matter for the mechanism.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard cross-country or pure aggregate approaches miss

- A standard cross-country finance-growth regression cannot distinguish whether finance causes productivity changes or merely forecasts them.
- A pure Hsieh–Klenow decomposition documents misallocation but does not explain **why** it changes over time.
- A purely aggregate story of capital inflows cannot say whether funds went to the most productive firms or just to the firms with the strongest balance sheets.

4.2 What identifies the paper's mechanism

- **Step 1: document a very specific empirical pattern.** The rise in dispersion is in MRPK, not in MRPL. This points directly toward a capital-allocation story rather than a generic rise in wedges affecting all factors.
- **Step 2: connect dispersion to capital-productivity comovement.** The fall in the correlation between log productivity and log capital is exactly the pattern expected under capital misallocation.
- **Step 3: calibrate the model to micro moments.** The leverage-size gradient and within-firm capital dynamics are central to pinning down the size-dependent borrowing constraint.
- **Step 4: compare nested models.** The paper shows that the homogeneous-friction model and the no-friction model fail in systematic ways. This makes the size-dependent friction assumption empirically meaningful rather than cosmetic.

4.3 Why size dependence is the key identifying feature

- Without size dependence, lower real interest rates do not create enough heterogeneity in firm responses. Either too many firms stay constrained (HoF) or too many firms respond symmetrically (NoF).
- With size dependence, some firms become effectively unconstrained as they grow, while others remain constrained for a long time.
- That asymmetry is what produces the rising MRPK dispersion, declining productivity-capital correlation, and fall in allocative efficiency.

4.4 Direct evidence on the mechanism

- In the data, a firm's capital growth from 1999 to 2007 is positively related to its **initial net worth**, even after conditioning on its initial productivity and initial capital.
- The estimated elasticity in the data is about **0.17**; the baseline HeF model generates a similar elasticity of about **0.21**.
- The same relationship is much weaker in the nested models, especially in the no-frictions model.

Interpretation. This is powerful because it directly speaks to the paper’s mechanism: firms with more internal funds expand more, conditional on productivity.

4.5 Large-firm evidence

- The model predicts that the rise in MRPK dispersion should come mainly from differences **between small and large firms**, not from additional dispersion **within large firms**.
- The data confirm this: among firms in the top tail of the capital distribution, MRPK dispersion does not show the same rising trend.

Interpretation. This is a strong piece of out-of-sample evidence for the size-dependent borrowing-constraint channel.

4.6 Alternative explanations and why they fail

- **Financial liberalization as looser borrowing constraints.** In the model, a relaxation of borrowing constraints raises capital and borrowing but *improves* allocation, lowers MRPK dispersion, and raises TFP relative to efficient. That is the wrong sign for the Spanish pre-2008 facts.
- **A decline in uncertainty.** The dispersion of firm productivity shocks actually falls before the crisis. In the model, that too would reduce MRPK dispersion and improve allocation.
- **Post-crisis developments.** After 2008, rising uncertainty and tighter borrowing constraints can account for the further rise in MRPK dispersion and the reversal of capital flows.

4.7 Relation to the literature

- Relative to **Hsieh and Klenow (2009)**, this paper adds a dynamic, balance-sheet-based explanation for changing misallocation over time.
- Relative to **Midrigan and Xu (2014)** and **Moll (2014)**, the paper emphasizes that a decline in real interest rates can lower productivity during the transition, especially when borrowing constraints depend on firm size.
- Relative to standard finance-growth work, the paper opens the black box by connecting aggregate outcomes to **firm-level borrowing, net worth, and capital allocation**.

5 Tables and figures

5.1 Table I and Figure I: coverage and aggregate representativeness

Read:

- Table I shows that ORBIS-AMADEUS covers a large share of Spanish manufacturing throughout the sample, especially for the wage bill and gross output.

TABLE I
COVERAGE IN ORBIS-AMADEUS RELATIVE TO EUROSTAT (SBS):
SPAIN MANUFACTURING

	Employment	Wage bill	Gross output
1999	0.56	0.69	0.75
2000	0.58	0.71	0.76
2001	0.61	0.73	0.77
2002	0.65	0.75	0.79
2003	0.65	0.74	0.78
2004	0.66	0.75	0.78
2005	0.66	0.74	0.77
2006	0.67	0.74	0.77
2007	0.67	0.74	0.77
2008	0.65	0.72	0.72
2009	0.71	0.72	0.75
2010	0.68	0.73	0.74
2011	0.69	0.74	0.75
2012	0.65	0.71	0.72

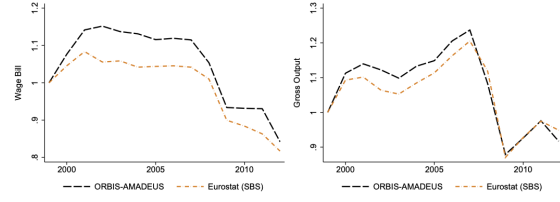


FIGURE I
Aggregates in ORBIS-AMADEUS and Eurostat (SBS)

TABLE II
SHARE OF TOTAL MANUFACTURING ECONOMIC ACTIVITY BY SIZE CLASS IN SPAIN (2006)

		Employment	Wage bill	Gross output
ORBIS-AMADEUS	1–19 employees	0.24	0.19	0.14
	20–249 employees	0.50	0.47	0.42
	250+ employees	0.26	0.34	0.45
Eurostat (SBS)	0–19 employees	0.31	0.20	0.14
	20–249 employees	0.43	0.43	0.38
	250+ employees	0.26	0.37	0.49

- Figure I shows that aggregate wage bill and gross output in the sample move closely with Eurostat aggregates.

Takeaway. The data are broad enough that the time-series patterns in the paper are unlikely to be driven by a tiny or highly selective sample of firms.

5.2 Table II: representativeness by firm size

Read:

- Small and medium-sized firms account for similar shares of employment, wage bill, and output in ORBIS-AMADEUS and Eurostat.
- This matters because the paper’s mechanism works through differences between smaller constrained firms and larger better-capitalized firms.

Takeaway. The sample is well suited for studying size-dependent financial frictions.

5.3 Figure II and Figure III: rising capital-return dispersion and TFPR moments

Read:

- Figure II shows a strong upward trend in the dispersion of log MRPK, but not in log MRPL.
- Figure III shows that TFPR dispersion rises and the covariance between TFPR and the capital-labor ratio becomes more negative over time.

Takeaway. The key reallocation problem is on the **capital** side, not the labor side.

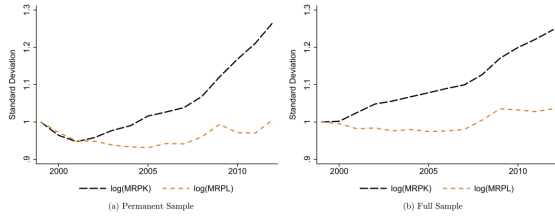


FIGURE II
Evolution of MRPK and MRPL Dispersion

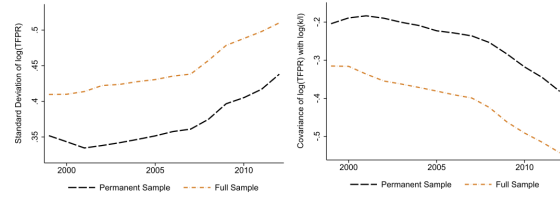


FIGURE III
TFPR Moments

5.4 Figure IV and Figure V: allocative efficiency and observed TFP

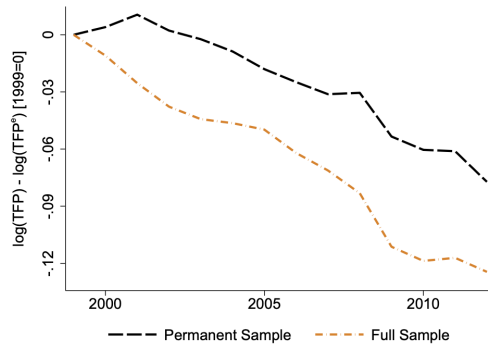


FIGURE IV
Evolution of TFP Relative to Efficient Level

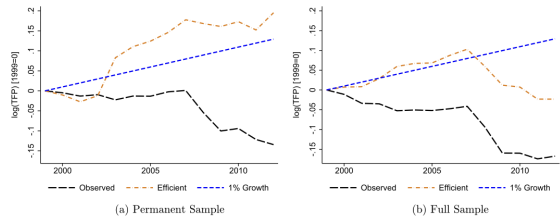


FIGURE V
Evolution of Observed TFP Relative to Benchmarks

Read:

- Figure IV plots $\log \text{TFP} - \log \text{TFP}^e$ and shows that actual TFP drifts farther below the efficient benchmark over time.
- Figure V compares observed TFP to benchmark paths and shows that observed productivity underperforms them.

Takeaway. Rising MRPK dispersion is not just a harmless change in cross-sectional prices; it is associated with a meaningful deterioration in allocative efficiency.

5.5 Figure VI: capital becomes less aligned with productivity

Read:

- The standard deviation of log capital rises steadily.
- The correlation between log capital and log productivity declines.

Takeaway. Capital inflows appear to be going disproportionately to less productive firms, exactly the pattern the model seeks to explain.

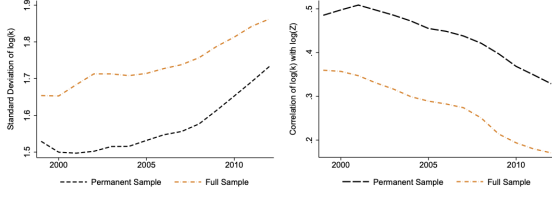


FIGURE VI
Log Capital Moments

TABLE III
BASELINE PARAMETERS

β	γ	δ	ε	α	ρ	σ	π	z_L	z_H	ψ	λ_0	λ_1
0.87	2.00	0.06	3.00	0.35	0.59	0.13	0.80	0.785	1.86	3.20	0.98	0.047

TABLE IV
FIRM-LEVEL MOMENTS: MODEL VS. DATA (1999–2007)

Moment	Data	HeF	HoF	NoF
Panel A: Distributional moments				
1. Std.dev.(log Z)	0.37	0.38	0.38	0.38
2. Std.dev.(log ℓ)	1.13	0.80	0.74	0.71
3. Std.dev.(log k)	1.52	0.91	0.75	0.66
4. Top 20% share of aggregate labor	0.69	0.63	0.59	0.58
5. Top 20% share of aggregate capital	0.79	0.71	0.61	0.56
Panel B: Within-firm moments				
6. Coefficient of $\frac{k-k}{k-k}$ on log Z	0.10	0.10	0.10	0.10
7. Coefficient of $\frac{k-k}{k-k}$ on log a	0.09	0.04	0.27	0.01
8. Coefficient of $\frac{k-k}{k-k}$ on log k	-0.46	-0.07	-0.46	-0.19
9. Coefficient of $\frac{b-b}{k-k}$ on log Z	-0.38	-0.04	-0.06	0.03
10. Coefficient of $\frac{b-b}{k-k}$ on log a	0.15	0.09	0.72	0.02
11. Coefficient of $\frac{b-b}{k-k}$ on log k	-0.34	0.07	-0.74	-0.22
Panel C: Cross-sectional moments				
12. Fraction borrowing	0.90	0.90	0.90	1.00
13. Coefficient of $\frac{b}{k}$ on log k	0.15	0.15	0.03	0.00
14. Corr (log Z, log k)	0.62	0.88	0.89	0.91
15. Corr (log Z, log a)	0.75	0.84	0.90	0.43
16. Corr (log MRPK, log Z)	0.03	-0.08	0.28	0.51
17. Corr (log MRPK, log k)	-0.62	-0.54	-0.18	0.10
18. Corr (log MRPK, log a)	-0.14	-0.36	-0.15	0.20
Panel D: Model evaluation				
19. Root mean squared error		0.28	0.35	0.41
20. Mean absolute error		0.22	0.29	0.34

5.6 Table III: baseline parameters

Read:

- The conventional macro parameters are fairly standard: $\beta = 0.87$, $\gamma = 2$, $\delta = 0.06$, $\varepsilon = 3$, $\alpha = 0.35$.
- The key calibrated friction parameters are $\psi = 3.20$, $\lambda_0 = 0.98$, and $\lambda_1 = 0.047$.

Takeaway. The model does not rely on an exotic calibration. The nonstandard part is specifically the **size dependence** of borrowing capacity.

5.7 Table IV: firm-level moments in the data and in the model

Read:

- The HeF model matches the targeted moments by construction, but it also does better on untargted moments than the HoF and NoF models.
- In particular, it gets much closer to the observed role of net worth in capital and debt dynamics and to the observed negative relation between MRPK and firm size.

- The root mean squared error is lowest in the heterogeneous-frictions model.

Takeaway. The micro evidence strongly favors a model in which borrowing capacity increases with firm size.

5.8 Table V and Figure VIII: aggregate responses in the baseline model

	Data	Model HeF
1. $\Delta \Lambda$	−0.031	−0.023
2. Δ (Dispersion)	0.034	0.074
3. $\Delta \log(\sum y)$	0.14	0.26
4. $\Delta \log K$	0.22	0.38
5. $\frac{\Delta B}{\Delta K}$	1.28	1.24

Notes. All changes are calculated between 1999 and 2007. $\Delta \Lambda$ denotes the percentage point change in TFP relative to its efficient level, that is the difference between $\log(TFP_{07}) - \log(TFP_{99})$ and $\log(TFP_{99}) - \log(TFP_{99}^e)$. Δ (Dispersion) is the $\Delta[\text{std. dev.}(\log(MRPK))]$. $\Delta \log(\sum y)$ is the change in the log of the sum of output and $\Delta \log K$ is the change in the log of aggregate capital. $\frac{\Delta B}{\Delta K}$ is the change in aggregate debt between 1999 and 2007 divided by the change in the aggregate capital between 1999 and 2007.

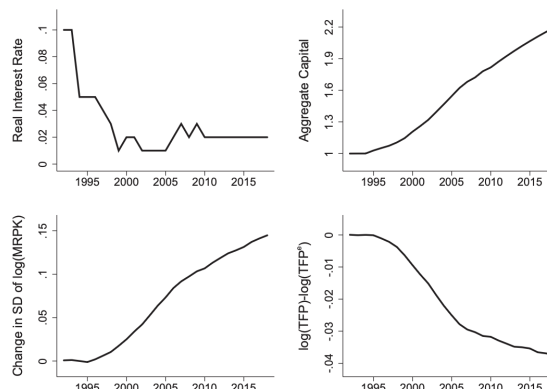


FIGURE VIII
Evolution of Aggregate Variables in Baseline HeF Model

Read:

- In the data between 1999 and 2007, TFP relative to efficient falls by about **0.031**, MRPK dispersion rises by about **0.034**, output rises by about **0.14**, and capital rises by about **0.22**.
- The baseline model generates the same qualitative pattern and about three-quarters of the decline in allocative efficiency.
- Debt rises more than one-for-one with capital in both the data and the model.

Takeaway. The model does not just fit micro moments; it also reproduces the main aggregate pattern the paper is trying to understand.

5.9 Figure IX and Figure X: the mechanism and the large-firm test

Read:

- Figure IX compares a constrained and an unconstrained firm after the real rate falls. The unconstrained firm accumulates capital and debt quickly and sees its MRPK fall. The constrained firm cannot keep up, so its MRPK stays high.
- Figure X shows that MRPK dispersion does not rise within the subsample of large firms, both in the data and in the model.

Takeaway. The key mechanism is not that every firm responds differently at random; it is that large, better-capitalized firms pull away from smaller constrained firms.

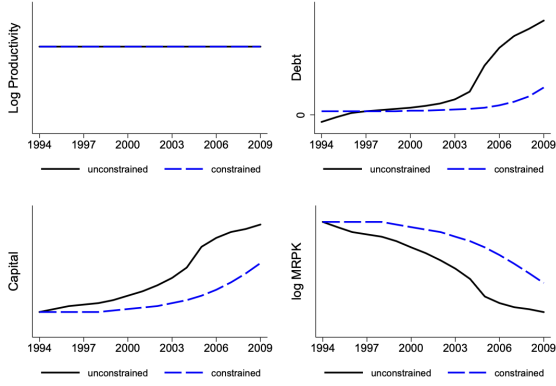


FIGURE IX

Real Interest Rates and Misallocation: An Illustrative Example (HeF Model)

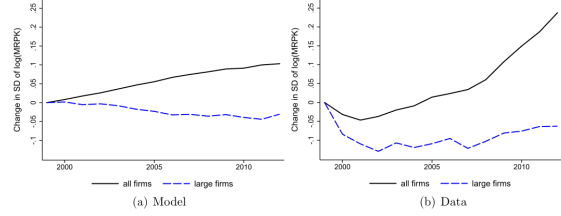


FIGURE X

MRPK Dispersion and Large Firms

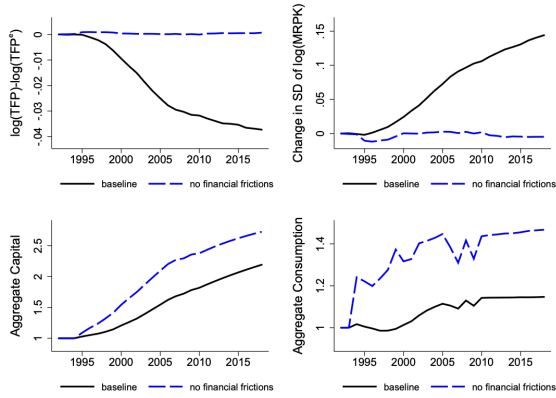


FIGURE XI

Impact of Misallocation on Evolution of Aggregate Variables

TABLE VI
AGGREGATE RESPONSES: DATA VS. VARIOUS MODELS (1999–2007)

	$\Delta\Lambda$	$\Delta(\text{Dispersion})$	$\Delta\log(\sum y)$	$\Delta\log K$	$\frac{\Delta\beta}{\Delta K}$
1. Data	-0.031	0.034	0.14	0.22	1.28
2. Heterogeneous frictions	-0.023	0.074	0.26	0.38	1.24
3. Homogeneous frictions	0.000	0.003	0.01	0.01	0.86
4. No frictions	0.000	0.002	0.27	0.43	1.06
5. Higher adjustment costs	-0.019	0.050	0.13	0.22	0.88
6. Exogenous labor wedges	-0.026	0.057	0.27	0.39	1.17
7. Overhead labor	-0.023	0.075	0.26	0.38	1.24
8. Unmeasured capital	-0.027	0.048	0.31	0.47	1.29
9. Alternative r process	-0.018	0.056	0.19	0.28	1.25

Notes. All changes are calculated between 1999 and 2007. $\Delta\Lambda$ denotes the percentage point change in TFP relative to its efficient level, that is the difference between $\log(\text{TFP}_{07}) - \log(\text{TFP}_{99}^*)$ and $\log(\text{TFP}_{99}) - \log(\text{TFP}_{99}^*)$. $\Delta(\text{Dispersion})$ is the $\Delta(\text{std. dev.}(\log(\text{MRPK})))$. $\Delta\log(\sum y)$ is the change in the log of the sum of output and $\Delta\log K$ is the change in the log of aggregate capital. $\frac{\Delta\beta}{\Delta K}$ is the change in aggregate debt between 1999 and 2007 divided by the change in the aggregate capital between 1999 and 2007.

TABLE VII
AGGREGATE RESPONSES AND VARIOUS SHOCKS

	$\frac{TFP}{TFP^*}$	Std. dev. (log MRPK)	K	$B - K$
1. Data (precrisis period)	↓	↑	↑	↑
2. Real interest rate decline ($\downarrow r$)	↓	↑	↑	↑
3. Financial liberalization ($\uparrow \lambda_0$ or $\uparrow \lambda_1$)	↑	↓	↑	↑
4. Uncertainty decline ($\downarrow \sigma$)	↑	↓	—	↑
5. Data (postcrisis period)	↓	↑	↓	↓

5.10 Figure XI, Table VI, and Table VII: model comparison and robustness

Read:

- Figure XI shows that removing financial frictions changes aggregate transition dynamics substantially: with no frictions, capital and output respond much more strongly and misallocation is absent.
- Table VI shows that the baseline HeF model is much more successful than HoF or NoF; robustness variants with exogenous labor wedges, overhead labor, unmeasured capital, or an alternative interest-rate process preserve the core result.
- Table VII shows that a pure relaxation of borrowing constraints or a decline in uncertainty predicts the wrong sign for the Spanish precrisis facts, while the fall in the real interest rate matches them qualitatively.

Takeaway. The paper’s narrative survives multiple robustness checks and is not easily mimicked by standard alternative shocks.

6 Short summary

- **Method.** Measure misallocation using Hsieh–Klenow-style firm-level factor-return dispersion, then explain its evolution with a dynamic model of heterogeneous firms, capital adjustment costs, and size-dependent financial frictions.
- **Identification.** The mechanism is pinned down by micro moments on borrowing, leverage, and capital accumulation, not just by aggregate fit.
- **Application.** Spanish manufacturing, 1999–2012, in the run-up to and aftermath of the euro-era capital inflow boom.
- **Main result.** The decline in the real interest rate can generate more capital, more debt, higher MRPK dispersion, and lower allocative efficiency because capital flows toward high-net-worth firms rather than the most productive firms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that Spanish manufacturing experienced a significant worsening in capital allocation between 1999 and 2012.

- It also establishes that a model with **size-dependent borrowing constraints** and a decline in the real interest rate can account for the main direction of those changes, both at the firm level and in the aggregate.
- A simple financial-liberalization story in which looser credit constraints improve allocation does not fit the Spanish precrisis evidence.

7.2 Economic intuition

- Lower real interest rates make capital accumulation more attractive for everyone.
- But financially stronger firms can act on this incentive immediately, while financially weaker firms must wait and self-finance.
- As richer firms expand first, they absorb a disproportionate share of the capital inflow. The result is a weaker alignment between capital and productivity, a rise in MRPK dispersion, and a fall in aggregate efficiency.

7.3 Takeaway

This paper's main lesson is that the macro effect of low interest rates depends on the **micro structure of finance**. In an economy with size-dependent financial frictions, cheaper capital can produce **more investment but worse allocation**. That is the core reason why capital inflows in South Europe could coexist with sluggish productivity growth.